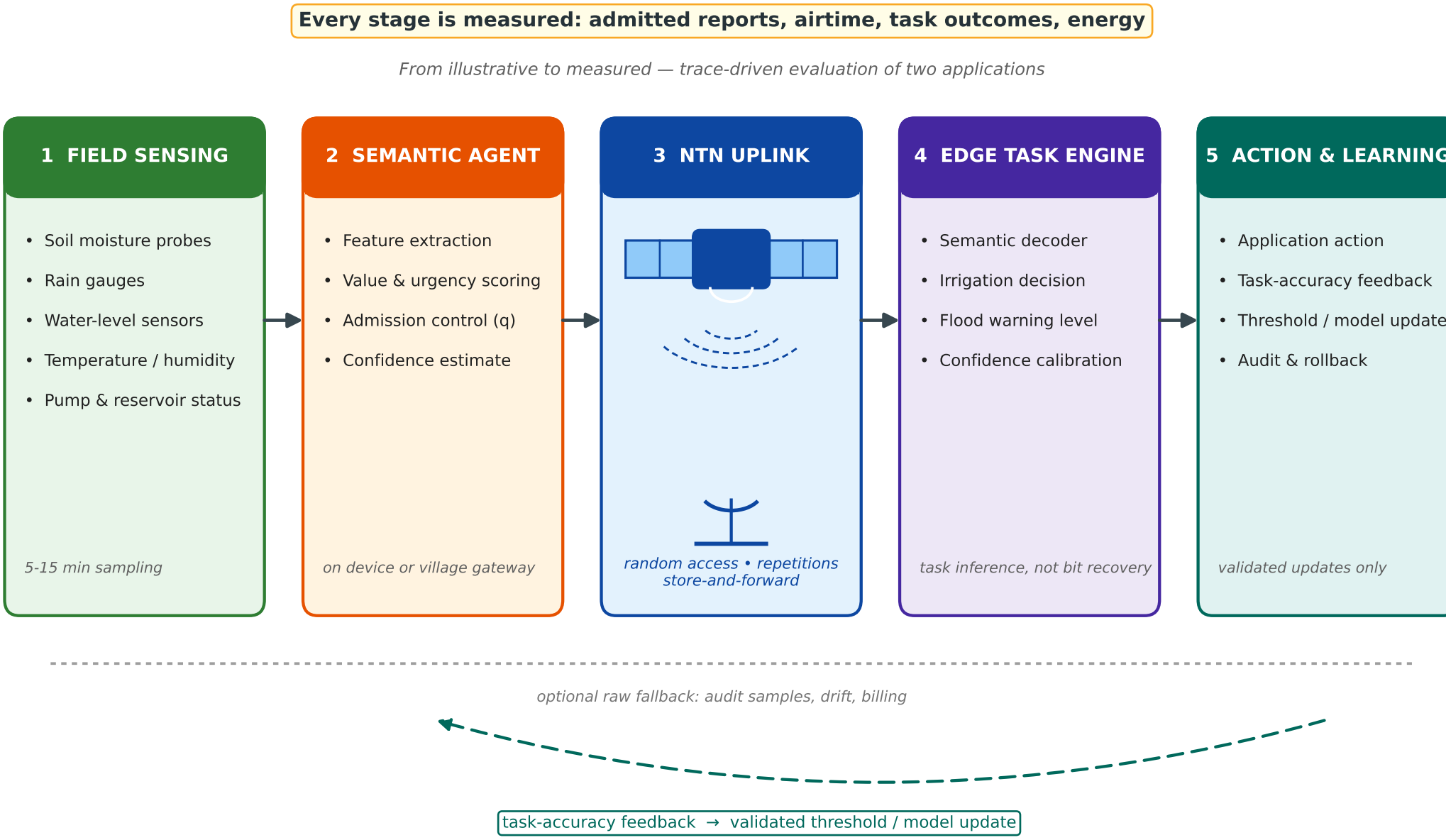

Figure 1. Application-specific semantic / task-oriented mIoT over NTN: field sensing, on-device semantic agent, LEO uplink, edge task engine, and guarded learning loop.